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Generating and Evaluating Synthetic UK Primary Care Data: Preserving Data Utility & Patient Privacy

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Abstract **– There is increasing interest in the potential of synthetic data to validate and benchmark machine learning algorithms as well as reveal any biases in real-world data used for algorithm development. This paper discusses the key requirements of synthetic data for such purposes and proposes an approach to generating and evaluating synthetic data that meets these requirements. We propose a framework to generate and evaluate synthetic data with the aim of simultaneously preserving the complexities of ground truth data in the synthetic data whilst also ensuring privacy. We include as a case study, a proof-of-concept synthetic dataset modelled on UK primary care data to demonstrate the application of this framework.**

Keywords-primary care; synthetic data; synthetic data generation; synthetic data evaluation

I. INTRODUCTION

The Clinical Practice Research Datalink (CPRD) is a UK primary care database covering 15 percent of the UK population and includes over 17,400 clinical event types across patients with 25% of the patient data tracing back at least 20 years. The recent development of intelligent applications makes these data attractive for the application of various data mining and machine learning algorithms to yield new insights into drug use patterns [3] and for diagnosis or prediction of diseases [1]. However, there is a need for developing a synthetic dataset that would complement such rich real-world data for various reasons outlined below.

1) Ease of access - Access to individual record level real-world data, even in pseudonymised format with strong personal identifiers removed, tends to be strictly regulated to control the risk of inadvertent patient re-identification [2].

2) Cost-efficiency - In the context of UK primary care data collection, using a synthetic data generation model for benchmarking and validation is significantly more cost-effective than expanding the population coverage of real-world data due to the cost of scaling-up collection and processing pipelines.

3) Test efficiency – Using a synthetic data generation model can efficiently improve algorithms or functions in an information system [4] by generating desired data on-the-fly.

4) Patient privacy protection - the social-demographic and health-related content in the primary care data makes patient identification more likely and therefore a fully synthetic approach can effectively avoid this problem [17].

5) Completeness - It can be difficult to conduct unbiased primary care data research if there are inherent biases in the data (either longitudinally or latitudinally). Synthetic data can supplement real data by either filling gaps or enlarging a sub-group dataset [16]. In addition, anonymisation measures for real data may compromise data utility due to information loss [19].

6) Benchmarking capabilities - This is useful when comparing different machine learning methods against a standardized dataset while focussing on a specific set of diseases (e.g. cardiovascular diseases).

Despite all the advantages outlined above, there is no synthetic data generation and evaluation approach that can be applied to UK primary care data to ensure that the generated data preserve the ground truth (such as sensible biological relationships between variables) whilst ensuring privacy.

In this paper, we propose a framework to enable synthetic primary care data generation ensuring similarities to ground truth and the preservation of patient privacy. The following four key requirements are envisioned to be critical for making synthetic primary care data usable:

Preservation of biological relationships. The chosen data variables and target studies should preserve the correct underlying relationships (e.g. female specific diseases should not include male patients) or well-established clinical symptom-diagnosis pairs should be preserved (e.g. excessive thirst because of diabetes).

Univariate similarity. Each concerned variable in the synthetic data should have similar fundamental aggregated statistics to the ground truth (e.g. similar distributions) for both continuous variables and categorical variables.

Multivariate similarity. Data with multiple dimensions often have correlated structure between data variables. Retaining such correlational structure in synthetic data is crucial to ensure that it is a truthful representation of the real world. In some studies, initial knowledge (for e.g. clinical expertise or familiarity with real world distributions) may be required to inform the development but this would only be feasible when the number of concerned variables is relatively small. However, when the variable numbers are large such as in [1], a manual approach will be a challenge.

Preservation of patient privacy. Privacy might be a concern even in the case of fully synthetic data. This can occur when synthetic generation produces a very similar dataset in terms of aggregated characters to real-world data. For example, a small number (i) of rare disease cases in the

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synthetic data occur in the same region in patients with similar ages. Thus, when i is sufficiently small, there is a chance that these outliers can be identified from the data. Essentially, this occurs when a synthetically generated patient shares the same characteristics as a real patient purely by chance. As a result, it is important to consider mechanisms which can be put in place to protect against such a scenario.

The paper is organized as follows. Section II reviews existing approaches and challenges to generating synthetic datasets, privacy concerns relating to generated data and multidimensional complexities in primary care data. Section III presents the proposed framework followed by a case study in Section IV, and section V concludes the paper.

II. RELATED WORKS

This section reviews previous work in synthetic data generation, privacy preservation and complexity of healthcare data to inform our proposed framework.

A. *Synthetic Data Generation Methods*

In general, synthetic data generation methods can be categorised into three groups:

Group 1: generating synthetic data based upon some statistical properties of the real-world data. This approach is useful when real data are difficult to access e.g. the data are transient and difficult to collect, or the data is scarce such as in the case of rare diseases or the distribution of events are highly imbalanced such as in the case of outliers [7]. In [14], data are sampled from the population distribution. In another practical example, the velocity property is used to generate synthetic data for various tactical moving objects in the military context [15].

Group 2: adding noise. Based upon adding noise to a small sample of data. This approach is particularly useful when the data are sensitive and only part of the real-world data needs to be generated [5], [10] and [6]. A relevant technique often used is data imputation, which addresses missing values in the data [8] and [9].

Group 3: using machine learning techniques to generate /extend the dataset using prediction and inference. These techniques can be applied to generate both semi-synthetic data and fully synthetic data. In [12], a hidden Markov model is used to discover the hidden properties of data and generate the semi-synthetic medical process data. In [13], generative models are used to create both semi and fully synthetic data in relational databases in replace of real data.

B. *Privacy Preservation*

A range of approaches to mitigate the risk of patient privacy disclosure such as perturbation, condensation, randomisation, and fuzzy based methods have been used [18] to eliminate links between identifiable data and the data subject [20], [21]. However, anonymization of data can compromise its utility [19] if important information is removed. In addition, the anonymised data may still present a residual risk to privacy as described in [11]. In [22], a robust algorithm is further proposed to de-anonymise large sparse datasets.

A fully synthetic dataset can effectively tackle the privacy issue in a sense that all data values can be completely different from real-world data. However, in the context of clinical studies it may be necessary to reserve various statistical and socio-demographic characteristics from real-world data. The risk of privacy disclosure might still exist when outliers in synthetic data are the same as those in real-world data by chance.

C. *Complexity in Primary Care Data*

Complexity can come from two levels: first, access to heterogenous data sources [25] and second, the complex relationships inherent in the data. The latter can be data instance relationships within the data model of different entities such as patient entity and clinical observation entity.

In [23], an example is given to demonstrate a multidimensional model and 9 requirements of data modelling on patient diagnosis episodes, e.g. commonly occurring many-to-many relationships between patients and diagnoses should be handled by the model. The complex biological relationships are derived from empirical studies and observations. For example, a predefined set of variables such as cardiovascular risk factors [24].

While synthetic data can address the access issue, learning the many-to-many relationships within and across the data entities remains a challenge so that the fidelity of the clinical information can be retained.

III. SYNTHETIC PRIMARY CARE DATA FRAMEWORK

A. *Overview*

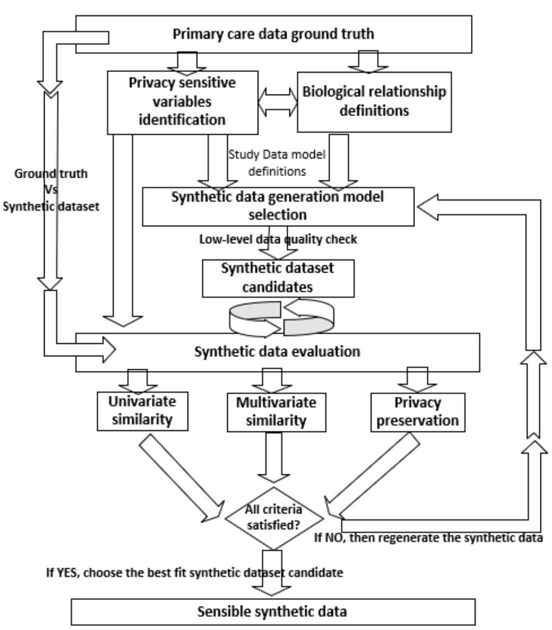

Figure 1 Synthetic primary care data framework overview

A synthetic primary care data framework with the objective of enabling a historic approach to the synthesis of primary care data while ensuring similarities between ground truth and the synthetic data as well as preserving privacy, biological relationships and biological values, is proposed. Although this framework targets UK primary care, it could be applied to other contexts.

The framework (see Figure 1) outlines a set of processes including the ground truth selection process as input, the synthetic data generation process, the evaluation process and ultimately the sensible synthetic data selection as output.

The ground truth selection process entails two linked tasks which are a privacy sensitive variable identification task and a biological relationship definition task. The results from these two tasks can contribute to the study data model definition, which in turn determines the suitable synthetic data generation model.

The synthetic data generation process aims to produce a set of synthetic data candidates and ensure the low-level data quality, e.g. data values and format of the resultant data.

The evaluation process focuses on assessing the three key criteria between the selected ground truth and the generated synthetic datasets which includes univariate similarity, multivariate similarity, and privacy preservation.

The sensible synthetic data selection is a recursive process which ends when all evaluation criteria are met.

B. Ground Truth Selection Process

At a high level, the ground truth is closely dependent on the documented studies. In this framework, we assume there is a panel of experts who can generate the relevant key variables. In this process, the selected variables need to go through two tasks: namely the privacy sensitive variable identification and the biological relation definition.

In most cases, each variable alone may not cause privacy issues, but rather, when a combination of variables (tuple) is presented. However, without knowing the distributions of these tuples it is usually impossible to identify them. Two alternative approaches can be used: either by referencing relevant official guidance and standards or second, or by using an expert panel to predefine a set of sensitive variables.

Once the individual sensitive variables are listed, the researchers can further define variable combinations among them that are deemed to be privacy sensitive. Examples of variables defined as sensitive patient data include familial relationships, marital and occupational status, sexuality, eligibility for social benefits etc.

Well-established biological relationships between variables and the study target also need to be defined at this stage. For example, the condition lupus can be relevant when studying hypertension.

C. Synthetic Data Generation Process

Based upon three approaches to generate the synthetic data, namely, #1 statistical properties driven, #2 adding noise, and #3 using machine learning algorithms, a 2x2 matrix in terms of data privacy and number of studied variables is proposed to guide the selection of the synthetic data generation model (see TABLE I).

TABLE I SYNTHETIC DATA GENERATION MODEL SELECTION GUIDE

	Low privacy sensitive	High privacy sensitive
Small no. of variables	1,2,3	1,3
Large no. of variables	1,3	3

Once the synthetic data are generated, a low-level data quality check may be required for variables with continuous values so that they are biologically plausible, e.g. blood pressure values in most cases should round to 0 decimal place.

D. Evaluation Process

As synthetic data generation models can produce a set of synthetic candidates, the evaluation process must at a high level differentiate these candidates in terms of their distance to the ground truth via certain distance based functions $dis(X^s, X^g)$, where the X^s and X^g represent the input spaces of synthetic data and ground truth respectively for some evaluation criterion. In the proposed framework, four criteria are defined:

1) Univariate similarity

The similarity can be further defined by two types of variables, the discrete variables $\{d_1, \ldots, d_i\} \in D$ and continuous variables $\{c_1, \ldots, c_i\} \in C$. For the discrete variables, the probability value P is used to represent the individual variable; for the continuous variables, the value distribution function f is used. The univariate similarity sim_{un} can be defined as:

$$\boldsymbol{sim_{un}} = \boldsymbol{dis}(\boldsymbol{P}(\boldsymbol{d_i^s}), \boldsymbol{P}(\boldsymbol{d_i^g})) \text{ for discrete variables} \quad (1)$$

$$\boldsymbol{sim_{un}} = \boldsymbol{dis}(\boldsymbol{f}(\boldsymbol{d_i^s}), \boldsymbol{f}(\boldsymbol{d_i^g})) \text{ for continuous variables} \quad (2)$$

During the univariate similarity evaluation, each variable is independently assessed and the dis function needs to be consistent for a variable type. The results of the test, i.e. $\mathrm{sim_{un}}$, must be interpreted in the context of the chosen dis function. For example, if a simple probability value is used for discrete variables, then a threshold value representing the minimum tolerated difference between two datasets needs to be defined. If a hypothesis-based statistical approach e.g. Kolmogorov-Smirnov test (KS test) is chosen for continuous variables, then the p value should be used.

2) Multivariate similarity

The multivariate analysis is required to compare the inter-relationship and patterns of data instances within two datasets $\{\boldsymbol{s_1^s}, \ldots, \boldsymbol{s_i^s}\} \in \boldsymbol{S^s}$ for synthethic dataset and $\{\boldsymbol{s_1^g}, \ldots, \boldsymbol{s_i^g}\} \in \boldsymbol{S^g}$ for ground truth dataset.

The purpose of the dis function in the multivariate similarity test is to measure the Euclidian distance in a, possibly transformed, input space $X \in \mathbb{R}^k$, where n variables of a data instance can be mapped to shared k dimensions ($0 < k < n$) via a function $f_n^k(X)$ and at the same time preserving their representative value in their original space $\mathbb{R}^n$. When k is small, e.g. 2, a graphical approach can be used to illustrate the similarities if the dataset is large. Existing multivariate analysis methods such as ordination techniques including principle component analysis (PCA), or non-metric multidimensional scaling (NMDS) [29] can be used. The similarity between two datasets sim_{mu} thus can be defined as:

$$\boldsymbol{sim_{mu}} = \boldsymbol{dis}\left(\boldsymbol{f_n^k}(\boldsymbol{S^s} \in \mathbb{R}^{\boldsymbol{n}}), \boldsymbol{f_n^k}(\boldsymbol{S^k} \in \mathbb{R}^{\boldsymbol{n}})\right) \quad (3)$$

In addition, the dis function can also be a pair-based comparison measurement. A correlation matrix hence can be used to compare ground truth and synthetic data. Let the matrix $M^g = \left[br_{ij}^g\right]$, where br_{ij} is the correlation between i^{th} and j^{th} variables in ground truth. As a result, given the calculated correlation matrix $M^s = \left[br_{ij}^s\right]$ for the synthetic dataset. The distance function can be also defined as

$$\boldsymbol{br_s} = \boldsymbol{dis}\left(\boldsymbol{br_{ij}^s}\right) = \begin{cases} \boldsymbol{0} \;\; \boldsymbol{br_{ij}^s} \neq \boldsymbol{br_{ij}^g} \\ \boldsymbol{1} \;\; \boldsymbol{br_{ij}^s} \approx \boldsymbol{br_{ij}^g} \end{cases} \quad (4)$$

The researchers need to define ≈ (almost equal) or ≠ (not equal), in the form of some threshold value.

3) Privacy preservation

Privacy preservation is needed when both datasets have same data instances by chance for outliers. As a result, the sensitive variables or their possible combinations identified in the ground truth selection process form the input spaces. For a set of sensitive variables in the synthetic dataset, $S_k^s = \{s_1, \ldots, s_k\} \in \mathbb{R}^k$,where $1 \leq k \leq$ total number of variables for a data instance.

The objective of the distance function here is calculate the distances among targeting data instances of sensitive variable(s) in space $\mathbb{R}^k$, the ones with larger distance to the rest groups of the data instances are viewed as outliers $\boldsymbol{p_r^k}$.

$$P_r^k = {\scriptstyle argmax}\left(dis\left(S_k^{s\prime}, \widehat{S_k^s}\right)\right), S_k^{s\prime} \neq \widehat{S_k^s}, S_k^{s\prime} \cup \widehat{S_k^s} = S_k^s \quad (5)$$

Decision tree-based approaches, such as DBSCAN [28], can be used to in this test. Once the P_r^k in synthetic data, and $P_r^{k\prime}$ in ground truth are identified, then an exhaustive comparison can be carried out to test if same data instances exist. If no identical instances are found, then the privacy can be preserved.

E. Sensible Synthetic Data Selection Process

Sensible synthetic data are datasets that meet all the evaluation criteria. In order to achieve this, the following pseudocode (see TABLE II) is provided to enable this decision process.

TABLE II ALGORITHM TO SELECT THE SENSIBLE SYNTHETIC DATASET

Step 1. Initialisation
  Let S be the generated synthetic datasets,
  S' be the ground truth.
  S^c be the sensible synthetic data candidates
Step 2. Univariate comparison
  For each $s \in S$:
    If sim_{un} is satisfied, then add s to S^c
  If S^c is empty, then regenerate synthetic data, else go to next step.
Step 3. Multivariate comparison
  For each $s \in S^c$
    If sim_{mu} is satisfied, then keep s in S^c, else remove s from S^c
  If S^c is empty, then regenerate synthetic data, else go to next step.
Step 4. Privacy preservation
  For each $s \in S^c$
    If P_r^k are found in s, $P_r^{k\prime}$ are found in S'
      For each $p_r^k \in P_r^k$, $p_r^{k\prime} \in P_r^{k\prime}$
        If $p_r^k = p_r^{k\prime}$, then remove s from S^c, else keep s in S^c.
  If S^c is empty, then regenerate synthetic data, else go to next step.
Step 5. If S^c is not empty, then S^c will be the sensible synthetic dataset.

IV. A CASE STUDY IN CARDIOVASCULAR DISEASE

Here, we demonstrate how the proposed framework can be used to generate and evaluate synthetic data in the area of cardiovascular disease (CVD) research.

A. Ground Truth Selection

Fourteen common variables are used as risk factors of CVD (see TABLE III).

TABLE III VARIABLES USED IN CARDIOVASCULAR DISEASE STUDY

#	Variables	Type (Available Values)
1	Age	Numeric
2	Ethnicity	Categorical
3	Gender	Categorical (Male; Female)
4	BMI	Numeric (1 decimal place)
5	Cholesterol/HDL ratio	Numeric (1 decimal place)
6	Systolic blood pressure	Numeric (0 decimal place)
7	Diastolic blood pressure	Numeric (0 decimal place)
8	family history of CHD	Categorical (Yes; No)
9	Smoking status	Categorical (Current; Former; None)
10	Treated hypertension	Categorical (Yes; No)
11	Type 2 diabetes	Categorical (Yes; No)
12	Rheumatoid arthritis	Categorical (Yes; No)
13	Atrial fibrillation	Categorical (Yes; No)
14	Chronic kidney disease	Categorical (Yes; No)

For demonstration purposes, all variables are predefined as privacy sensitive variables because these variables in combination are thought to be more likely to identify a patient than when only a few of them are used. BMI as defined is only valid for patients over 16 years in this case, so child patients are not included.

A total of 96,709 data instances (patients), are used and key statistics for each variable are in line with existing publications using UK primary care data. For example, from a social-demographic perspective, the white population covers more than 98.2% of the whole population with 42.4% of female and 57.6% of male, and age ranges from 27 to 73 with a median of 53.

B. Using Bayesian Networks to Generate Synthetic Data

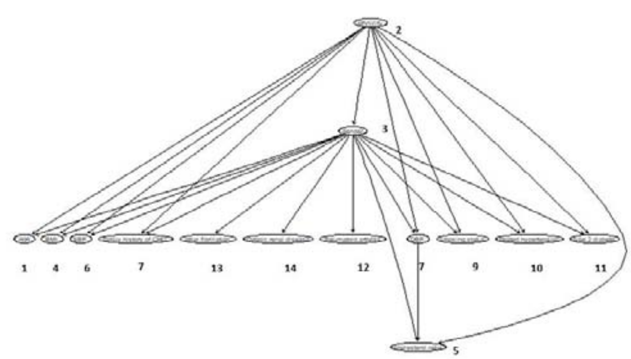

Figure 2 Inferred DAG of internal relationships between CVD variables (annotated by number from Table III) using BN structure learning

Bayesian networks (BNs) are a type of probabilistic graphical model that model a domain using conditional distributions and a graph structure. Inference algorithms can be used to learn graphical structures and to predict variable values. In the clinical study context, BNs are often used to discover new relationships among multiple factors [26] and to identify key factors that contribute to certain outcomes in CVD studies [27].

Figure 2 shows the inferred BN from the CVD data in the form of a directed acyclic graph (DAG) between variables. The accompanying conditional probabilities for each inter-linked variable are also learned which serve the basis for inferring new data, i.e. the synthetic data from the ground truth.

C. *Evaluation and Sensible Synthetic Dataset Selection*

The evaluation starts by sampling the same number of patients from both synthetic data and ground truth. In this case, a sample size of 800 is used i.e., one set of ground truth of 800 patients, and 5 sets of synthetic datasets (800 sample in each set) as the candidate sensible synthetic datasets. The following sections report the analysis processes of one of the candidate datasets that eventually meets all our criteria.

1) *Univariate similarity test*

For continuous variables, KS test is used (α =0.05, minimum p=0.53, maximum p=0.92). For discrete variables, the probabilities of available values are compared, and a probability difference smaller than 2% is acceptable in this case. Figure 3 shows two examples of the comparison methods used.

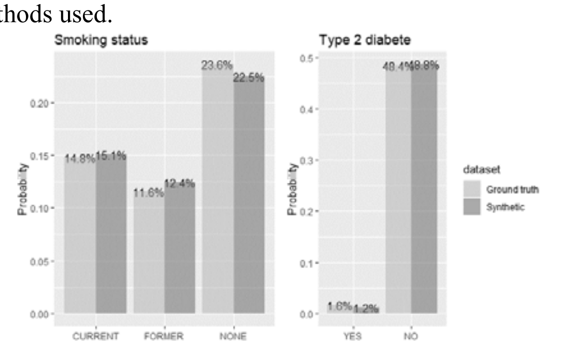

Figure 3 smoking status (left) and type 2 diabetes (right) univariate comparison in terms of probability

2) *Multivariate similarity test*

A correlation analysis is initially used to determine if the generated synthetic data captures the inter-relationships as would be expected in the ground truth. Correlations with p-values > 0.01 are considered as insignificant. Figure 4 confirms that one of the synthetic datasets can capture the significant correlations from ground truth. The correlation between gender and cholesterol ratio in this case is the only pair that is significantly correlated, i.e. 0.48 for synthetic data and 0.68 for ground truth.

In addition to the pair-based comparison, NMDS is also used here to explore how data instances are clustered in the multi-dimensional space. Data are mapped to 2-dimension space (stress =0.16) as shown in Figure 5.

Both datasets illustrate a clear similarity in terms of clusters and distribution. The similarity is further confirmed by an analysis of similarity (ANOSIM) (p=0.001 when α =0.01) which suggests that there is no significant difference between groups.

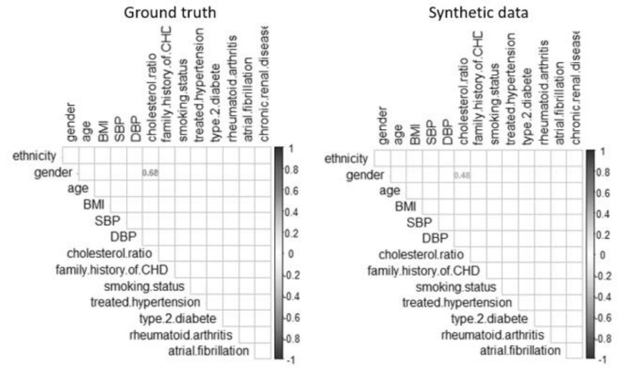

Figure 4 The correlation matrix for both ground truth (left) and synthetic data (right), all other the insignificant correlations are removed (p>0.01)

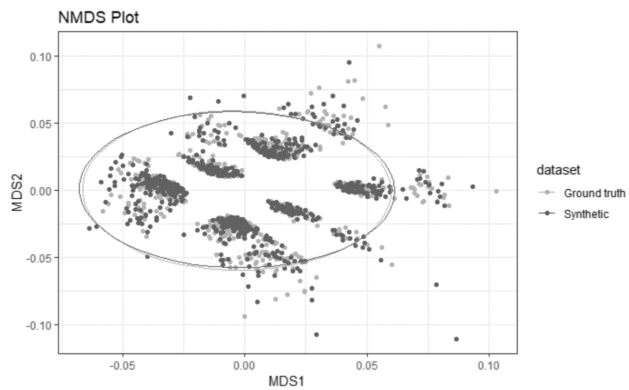

Figure 5 Representation of datasets in 2-dimension space using NMDS

3) *Outliers detection and comparison*

With the DBSCAN approach to cluster the dataset, 40 outliers are identified from both ground truth and synthetic data respectively (darkest points in Figure 6). A further exhaustive comparison between two outlier sets found that there is no duplication. This result suggests that the generated synthetic dataset is sensible.

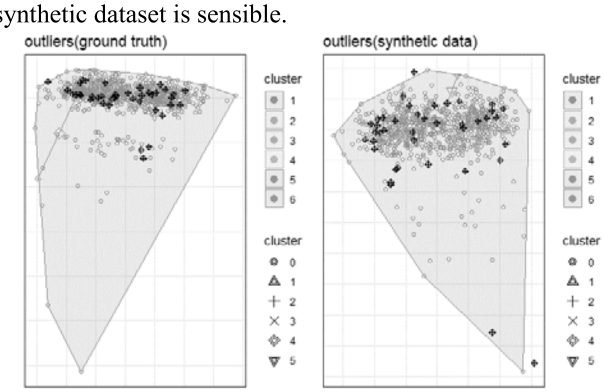

Figure 6 The detected outliers in both ground truth (40 data instances) and generated synthetic data (40 data instances).

V. CONCLUSION

A framework to generate and evaluate synthetic data for benchmarking purposes is proposed that addresses challenges in balancing synthetic data utility and preserving patient privacy. It covers the full lifecycle of the synthetic primary care data production from ground truth selection to synthetic data generation, evaluation and ultimately sensible synthetic data output.

The case study demonstrates that synthetic primary care data can be successfully generated and evaluated for CVD studies following the processes defined in the framework. The results provide a successful proof of concept which can be extended to other contexts.

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